Title: Progress Test 1

Course: AERO40006 (Ordinary Differential Equations)

Lecturer: Peter Vincent

**Question 1:** Identify the order of the following ODEs, also identify whether they are linear or non-linear. **(4 marks)**

(a) $e^{2x}\dfrac{\mathrm{d}^3y}{\mathrm{d}x^3} + \sin(x) = y$

(b) $\dfrac{\mathrm{d}y}{\mathrm{d}x} + y^4 = 0$

(c) $\cos(x)\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} - \sin(x)\dfrac{\mathrm{d}y}{\mathrm{d}x} + y^2 = y$

(d) $x\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} - 1 = \dfrac{\mathrm{d}^5y}{\mathrm{d}x^5}$

(a) linear, 3rd order

non-linear

(b) linear, 1st order

non-linear

(c) linear, 2nd order

(d) linear, 5th order

**Question 2:** Find particular solutions of the following first-order ODEs **(15 marks)**

(a) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{x^2+y^2}{xy} \qquad y(1) = 1$

(b) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = (1+y)(1+x) \qquad y(1) = 1$

(c) $(x-2)\dfrac{\mathrm{d}y}{\mathrm{d}x} - y = (x-2)^3 \qquad y(1) = 1$

(a) $\frac{dy}{dx} = \frac{x}{y} + \frac{y}{x}$

Let $u = \frac{y}{x}$ $\quad \therefore y = ux \quad \therefore \frac{dy}{dx} = u + \frac{du}{dx}x$

$\therefore \frac{dy}{dx} = u + \frac{1}{u} = u + \frac{du}{dx}x$

$\therefore \frac{du}{dx}x = \frac{1}{u}$

$\therefore \int u\,du = \int \frac{1}{x}\,dx$

$\therefore \frac{1}{2}u^2 - \ln|x| + C = 0$

$\therefore \frac{1}{2}\frac{y^2}{x^2} - \ln|x| + C = 0$

When $y(1) = 1$

$C = -\frac{1}{2}$

$\therefore \frac{y^2}{x^2} - 2\ln|x| - 1 = 0$

$\therefore y^2 = 2x^2\ln|x| + x^2$

$\therefore y = x\sqrt{2\ln|x| + 1}$

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(c) $(x-2)\dfrac{\mathrm{d}y}{\mathrm{d}x} - y = (x-2)^3 \qquad y(1) = 1$

(b) $\int \frac{1}{1+y}\,dy = \int 1+x\,dx$

$\therefore \ln|1+y| - x - \frac{1}{2}x^2 + C = 0$

When $y(1) = 1$

$\ln 2 - 1 - \frac{1}{2} + C = 0$

$\therefore C = \frac{3}{2} - \ln 2$

$\therefore \ln|1+y| - x - \frac{1}{2}x^2 + \frac{3}{2} - \ln 2 = 0$

$1+y \qquad\qquad 2$

$\therefore |1+y| = \frac{1}{2} e^x e^{\frac{1}{2}x^2} e^{-\frac{3}{2}}$

$\therefore y = 2e^x e^{\frac{1}{2}x^2} e^{-\frac{3}{2}} - 1$

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(c) $(x-2)\dfrac{\mathrm{d}y}{\mathrm{d}x} - y = (x-2)^3 \qquad y(1) = 1$

(c) As $x > 2$

$$\frac{dy}{dx} - \frac{1}{x-2}y = (x-2)^2$$

$\therefore$ Let $I = e^{\int -\frac{1}{x-2}dx} = e^{-\ln|x-2|+C} = \dfrac{1}{x-2}$ $\qquad \therefore \dfrac{dI}{dx} = -\dfrac{1}{(x-2)^2}$

$$\therefore I\frac{dy}{dx} - \frac{I}{x-2}y = I(x-2)^2$$

$$\therefore I\frac{dy}{dx} + \frac{dI}{dx}y = I(x-2)^2$$

$$\therefore \frac{dIy}{dx} = x-2$$

$$\therefore \int dIy = \int x-2\,dx$$

$$\therefore Iy = \frac{1}{x-2}y = \frac{1}{2}x^2 - 2x + C$$

When $y(1) = 1$,

$C = -1 + \dfrac{3}{2} = \dfrac{1}{2}$ $\qquad \therefore \dfrac{1}{x-2}y = \dfrac{1}{2}x^2 - 2x + \dfrac{1}{2}$

$$\therefore y = \frac{1}{2}(x^2-4x+1)(x-2) = \frac{1}{2}x^3 - 3x^2 + \frac{9}{2}x - 1$$

**Question 3:** Find general solutions of the following second-order ODEs **(15 marks)**

(a) $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} + 8\dfrac{\mathrm{d}y}{\mathrm{d}x} + 16y = 0$

(b) $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} - 7\dfrac{\mathrm{d}y}{\mathrm{d}x} + 12y = 3e^{3x}$

(c) $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4\dfrac{\mathrm{d}y}{\mathrm{d}x} + 9y = x$

(a) $\alpha^2 + 8\alpha + 16 = 0$

$\therefore\ b^2 - 4ac = 0$

$\therefore\ \alpha = -4$

$\therefore\ y = (A + Bx)e^{-4x}$

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(c) $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4\dfrac{\mathrm{d}y}{\mathrm{d}x} + 9y = x$

(b) $y = y_h + y_p$

For $y_h$,

$\alpha^2 - 7\alpha + 12 = 0$

$b^2 - 4ac = 1 > 0 \qquad \therefore \alpha = 3$ or $4$

$\therefore y_h = Ae^{3x} + Be^{4x}$

For $y_p$, let $y_p = xAe^{3x}$

$\therefore y_p' = Ae^{3x} + 3xAe^{3x} \qquad \therefore y_p'' = 3Ae^{3x} + 3Ae^{3x} + 9xAe^{3x}$

$\therefore 12xAe^{3x} - 7Ae^{3x} - 21xAe^{3x} + 6Ae^{3x} + 9xAe^{3x} = 3e^{3x}$

$\therefore -7A + 6A = -A = 3 \qquad \therefore A = -3$

$\therefore y_p = -3xe^{3x}$

$\therefore y = Ae^{3x} + Be^{4x} - 3xe^{3x}$

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(c) $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4\dfrac{\mathrm{d}y}{\mathrm{d}x} + 9y = x$

(c) $y = y_h + y_p$

For $y_h$,

$\alpha^2 + 4\alpha + 9 = 0$

$\because b^2 - 4ac = -20 < 0$

$\therefore \lambda_1 = -\dfrac{b}{2a} = -2$

$\lambda_2 = \dfrac{-\sqrt{b^2-4ac}}{2a} = \sqrt{5}$

$\therefore y_h = e^{-2x}[A\cos(\sqrt{5}x) + B\sin(\sqrt{5}x)]$

For $y_p$, let $y_p = Cx + D$ $\quad \therefore y_p' = C \quad \therefore y_p'' = 0$

$\therefore 0 + 4C + 9Cx + 9D = x$

$\therefore C = \dfrac{1}{9}$, $D = -\dfrac{4}{81}$

$\therefore y_p = \dfrac{x}{9} - \dfrac{4}{81}$

$\therefore y = e^{-2x}[A\cos(\sqrt{5}x) + B\sin(\sqrt{5}x)] + \dfrac{x}{9} - \dfrac{4}{81}$

**Question 4:** Derive Laplace Transforms of the following functions **(8 marks)**

(a) $f(t) = t$

(b) $f(t) = \sinh(at)$

(a) $\frac{df(t)}{dt} = 1$ , $f(0) = 0$

$L\{\frac{df}{dt}\} = sL\{f(t)\} - f(0)$

$\therefore L\{1\} = sL\{t\}$

$\because L\{1\} = \int_0^\infty e^{-st} dt = [\frac{-e^{st}}{s}]_0^\infty = \frac{1}{s}$

$\therefore sL\{t\} = \frac{1}{s}$

$\therefore L\{t\} = \frac{1}{s^2}$

**Question 4:** Derive Laplace Transforms of the following functions **(8 marks)**

(a) $f(t) = t$

(b) $f(t) = \sinh(at)$

(b) $$\frac{df(t)}{dt} = \left(\frac{e^{at} - e^{-at}}{2}\right)' = a\,\frac{e^{at} + e^{-at}}{2} = a\cosh(at)$$

$$\frac{d^2f(t)}{dt^2} = a^2\,\frac{e^{at} - e^{-at}}{2} = a^2 \sinh(at)$$

$$f(0) = 0\,,\quad \frac{df}{dt}(0) = a$$

$$\because\; L\left\{\frac{df}{dt}\right\} = sL\{t\} - f(0)$$

$$\therefore\; L\left\{\frac{d^2f}{dt^2}\right\} = sL\left\{\frac{df}{dt}\right\} - \frac{df}{dt}(0) = s^2L\{t\} - sf(0) - \frac{df}{dt}(0)$$

$$\therefore\; L\{a^2\sinh(at)\} = s^2L\{\sinh(at)\} - 0 - a$$

$$\therefore\; aL\{\sinh(at)\} - s^2L\{\sinh(at)\} = -a$$

$$\therefore\; L\{\sinh(at)\} = \frac{a}{s^2 - a}$$

**Question 5:** Solve the following ODE using Laplace Transforms. Feel free to use the table in Fig. 1 **(8 marks)**

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 3\frac{\mathrm{d}y}{\mathrm{d}x} + 2y = 0 \qquad y(0) = 1 \qquad \frac{\mathrm{d}y}{\mathrm{d}x}(0) = -3$$

$$L\left\{\frac{d^2y}{dx^2}\right\} - 3L\left\{\frac{dy}{dx}\right\} + 2L\{y\} = 0$$

$$\therefore s^2L\{y\} - sy(0) - \frac{dy}{dx}(0) - 3sL\{y\} + 3y(0) + 2L\{y\} = 0$$

$$\therefore (s^2 - 3s + 2)L\{y\} = s - 3 - 3 = s - 6$$

$$\therefore L\{y\} = \frac{s-6}{s^2-3s+2} = \frac{s-6}{(s-1)(s-2)}$$

$$= \frac{5}{s-1} + \frac{-4}{s-2}$$

$$\therefore f(x) = L^{-1}\left\{\frac{5}{s-1} - \frac{4}{s-2}\right\}$$

$$= 5L^{-1}\left\{\frac{1}{s-1}\right\} - 4L^{-1}\left\{\frac{1}{s-2}\right\}$$

$$= 5e^{t} - 4e^{2t}$$

**Question 6:** This is a bonus question for zero marks! Solve the following ODE. I do not know what the answer is, or even if an analytical solution exists. The ODE arose during the course of my research (into how macromolecules attached to the internal surface of blood vessels respond to oscillatory blood flow). **(0 marks)**

$$\frac{\mathrm{d}y}{\mathrm{d}t} - \alpha y = \sin(y)[A + B\sin(\omega t)] \qquad\qquad y(0) = 0$$

**(Total Marks: 50)**

$$L\left\{\frac{dy}{dt}\right\} - \alpha L\{y\} = L\{\sin(y)[A + B\sin(\omega t)]\}$$

$$sL\{y\} - y(0) - \alpha L\{y\} = [A + B\sin(\omega t)]\, L\{\sin(y)\}$$

| $f(t) = L^{-1}\{F(s)\}$ | $F(s) = L\{f(t)\}$ | |
|---|---|---|
| $k$ | $\frac{k}{s}$ | $s > 0$ |
| $e^{-kt}$ | $\frac{1}{s+k}$ | $s > -k$ |
| $te^{-kt}$ | $\frac{1}{(s+k)^2}$ | $s > -k$ |
| $t$ | $\frac{1}{s^2}$ | $s > 0$ |
| $t^2$ | $\frac{2}{s^3}$ | $s > 0$ |
| $\sin kt$ | $\frac{k}{s^2+k^2}$ | $s^2 + k^2 > 0$ |
| $\cos kt$ | $\frac{s}{s^2+k^2}$ | $s^2 + k^2 > 0$ |

Figure 1: Table of useful Laplace Transforms